

First Principles Calculations of Hydrogen Diffusion in Palladium

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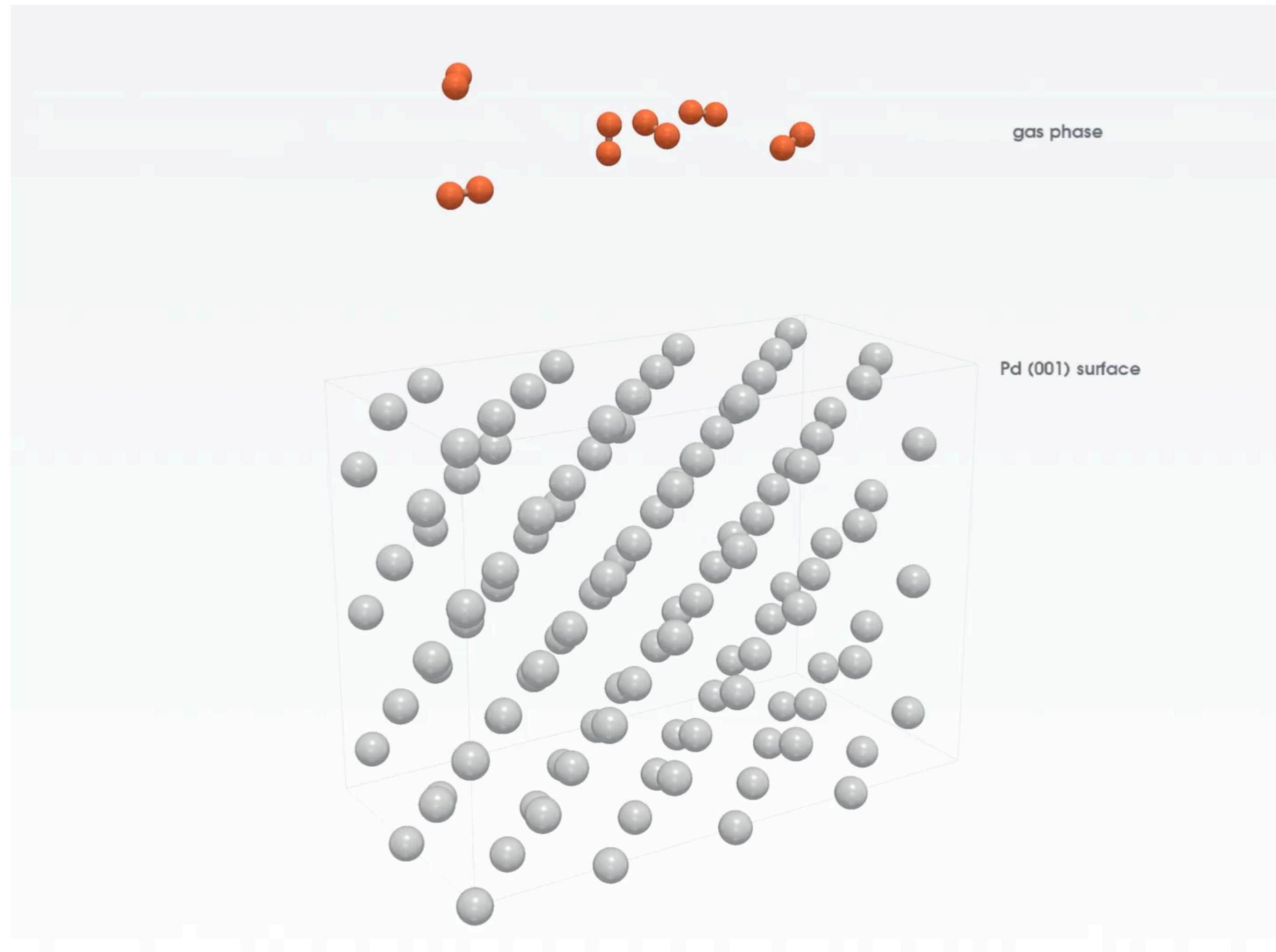
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Why Palladium? Why Diffusion?

- It efficiently dissociates H_2 at its surface and adsorbs the resulting H atoms into interstitial sites in its FCC lattice.
 $\Rightarrow$ This makes Pd important for hydrogen purification membranes, hydrogen sensors, hydrogen storage research, and catalytic hydrogenation.
- Atomic hydrogen moves fast through this metal: room temperature diffusion coefficient of $3.248 \times 10^{-11} \text{ m}^2/\text{s}$
- Our Goal: Predict this macroscopic diffusion rate from scratch using first principles calculations within the CP-PAW code

Source: Powell & Kirkpatrick, Phys. Rev. B 43, 6968–6976 (1991). DOI: 10.1103/PhysRevB.43.6968

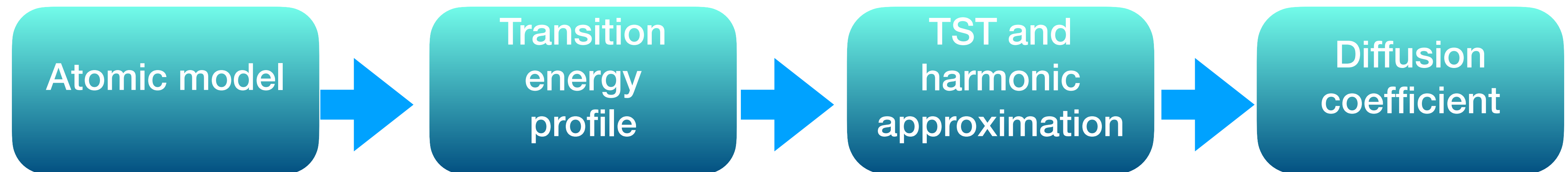
The Physical Mechanism of Hydrogen Diffusion in Palladium



- At Pd surface, H₂ dissociates and atomic H enters
- H preferentially occupies interstitial voids between Pd atoms
- Diffusion occurs through thermally activated jumps between sites
- We will investigate the energy transition within this jump

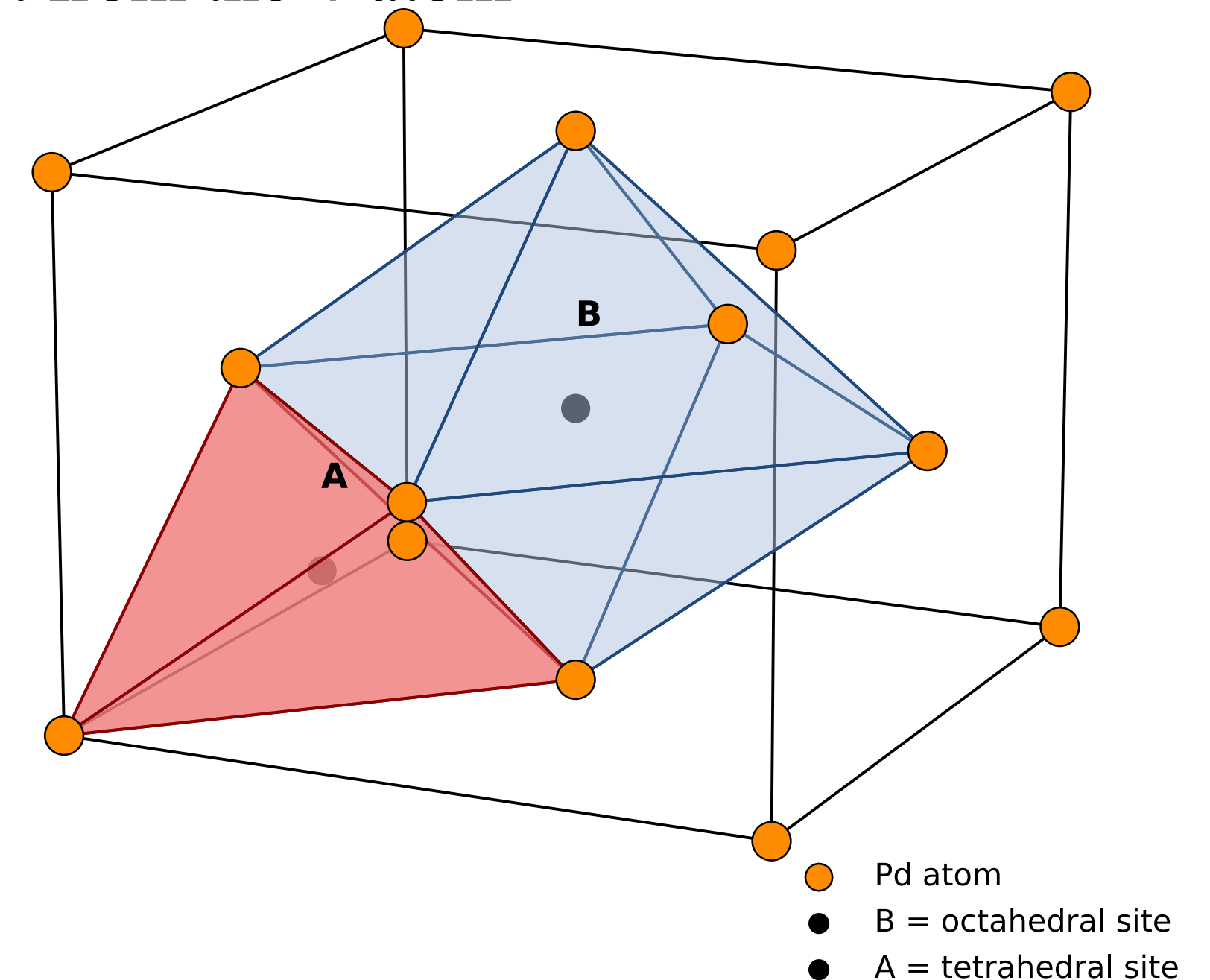
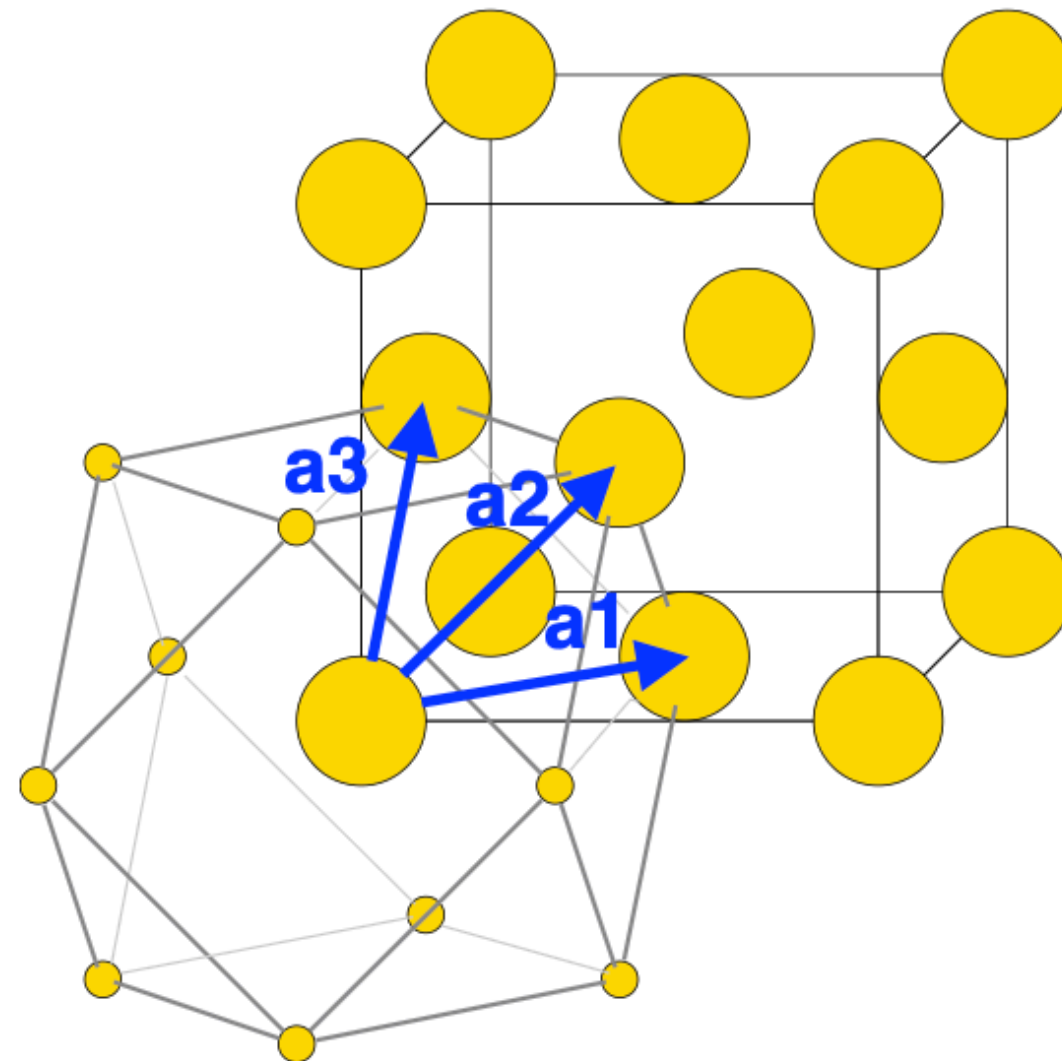
Computational Approach: Density Functional Theory

- Use the CP-PAW code to drive the system to the ground state and investigate the electronic structure from the perspective of a primitive unit cell
- Extend the system to a 32-atom supercell and insert one hydrogen atom
- Determine energies in the two stable configurations of H in Pd
- Drive the transition between these two configurations and determine energy profile
- Use Harmonic approximation and Transition-State Theory to obtain diffusion constant



Palladium

- Mass of $m \approx 106$
- Electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^0$ (5s electrons shift to the 4d orbital for maximum stability)
- Crystallizes in an fcc structure ($a_{\text{lat}}=3.89 \text{ \AA}$), which creates tetrahedral and octahedral voids
- Primitive unit cell holds a single atom with non-orthogonal lattice vectors (distinct from the 4-atom conventional cubic cell)



Source: CP-PAW Project Hydrogen in Palladium:Diffusion September 2, 2026

Electronic Structure of Palladium: Density of States

- Simulation in the primitive unit cell
- CP-PAW Parameters for metal: apply !MERMIN functional (T=0, TETRA+=T) to safely compute the varying occupancies
- Set SAFEORTHO=F to ensure the wave functions converge to true Hamiltonian eigenstates, avoiding artificial local minima

- States left of the Fermi level are occupied (valence band)

- States right of the Fermi level are unoccupied (conduction band)

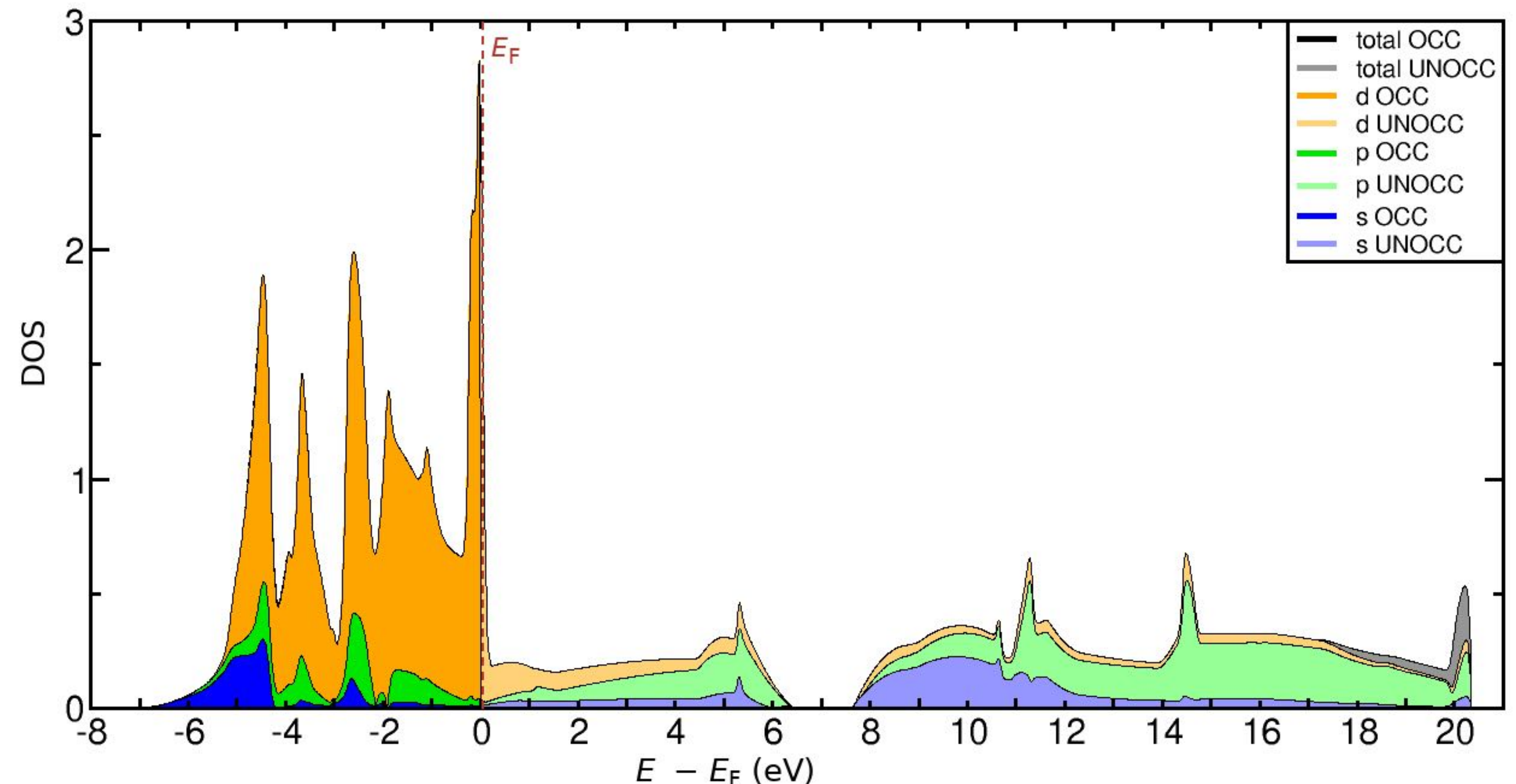
- d-Band dominance in the occupied states

- Fermi level is highest occupied energy state

⇒ Many electrons near E_F ready to participate in chemical reactions

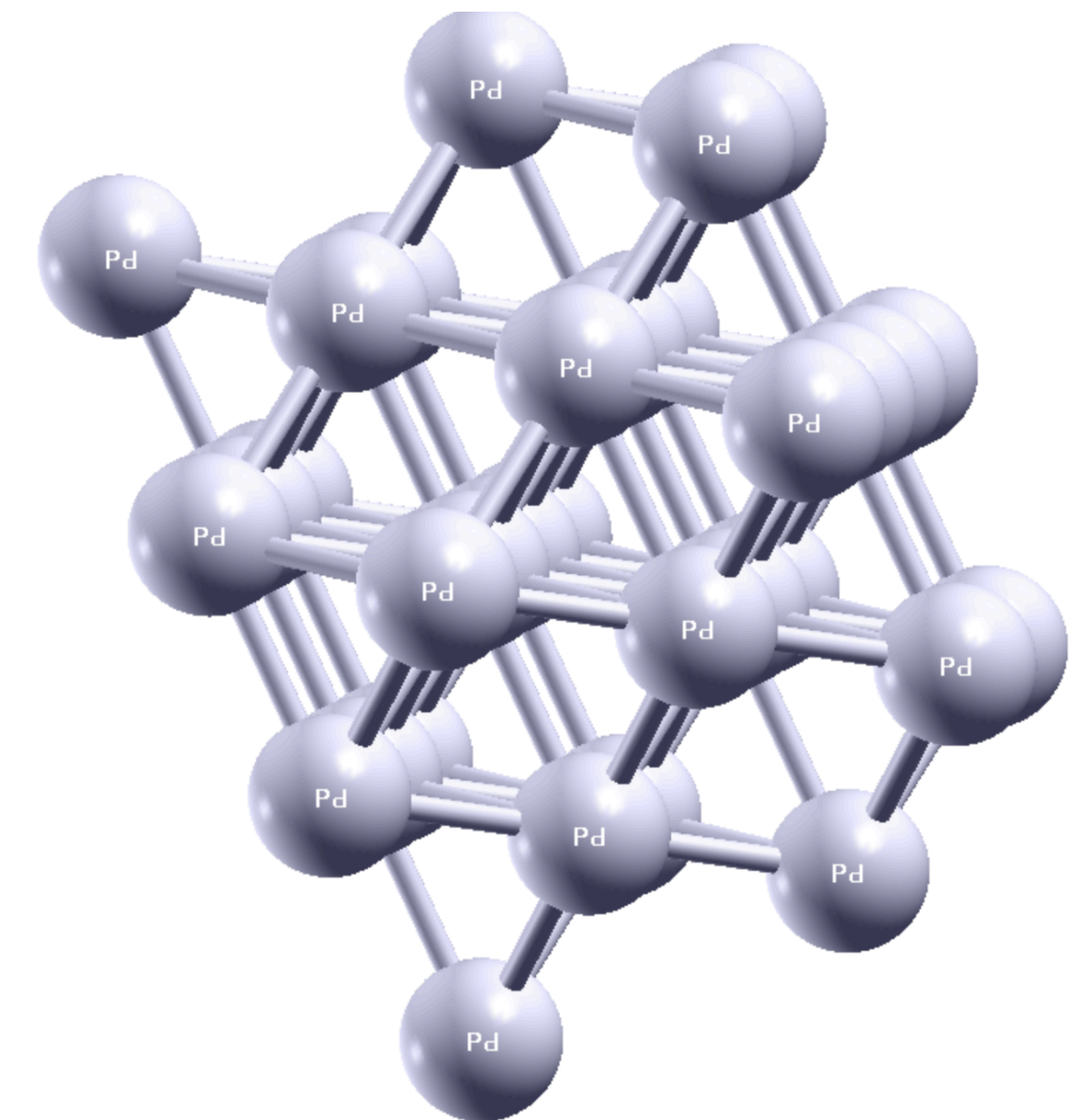
- Unoccupied regions: DOS flattens out

- Dominance of s and p contributions



Introduction of a 32-Atom Supercell

- Point defects like hydrogen impurities usually violate the translational symmetry of the host crystal
- Solve problem by studying a system with hydrogen atoms that are periodically repeated in the lattice
- Resulting unit cell is a supercell
- we consider a $2 \times 2 \times 2$ supercell (32 Pd atoms) which later hosts the hydrogen impurity
- Supercell must be so large that the hydrogen atoms do not experience each other
- 1 nm distance is sufficient
⇒ reciprocal unit cell and number of k-points shrink
- The latter is a measure for the accuracy
- Simulation shows: primitive unit cell has 63 k-points and supercell has 8 k-points



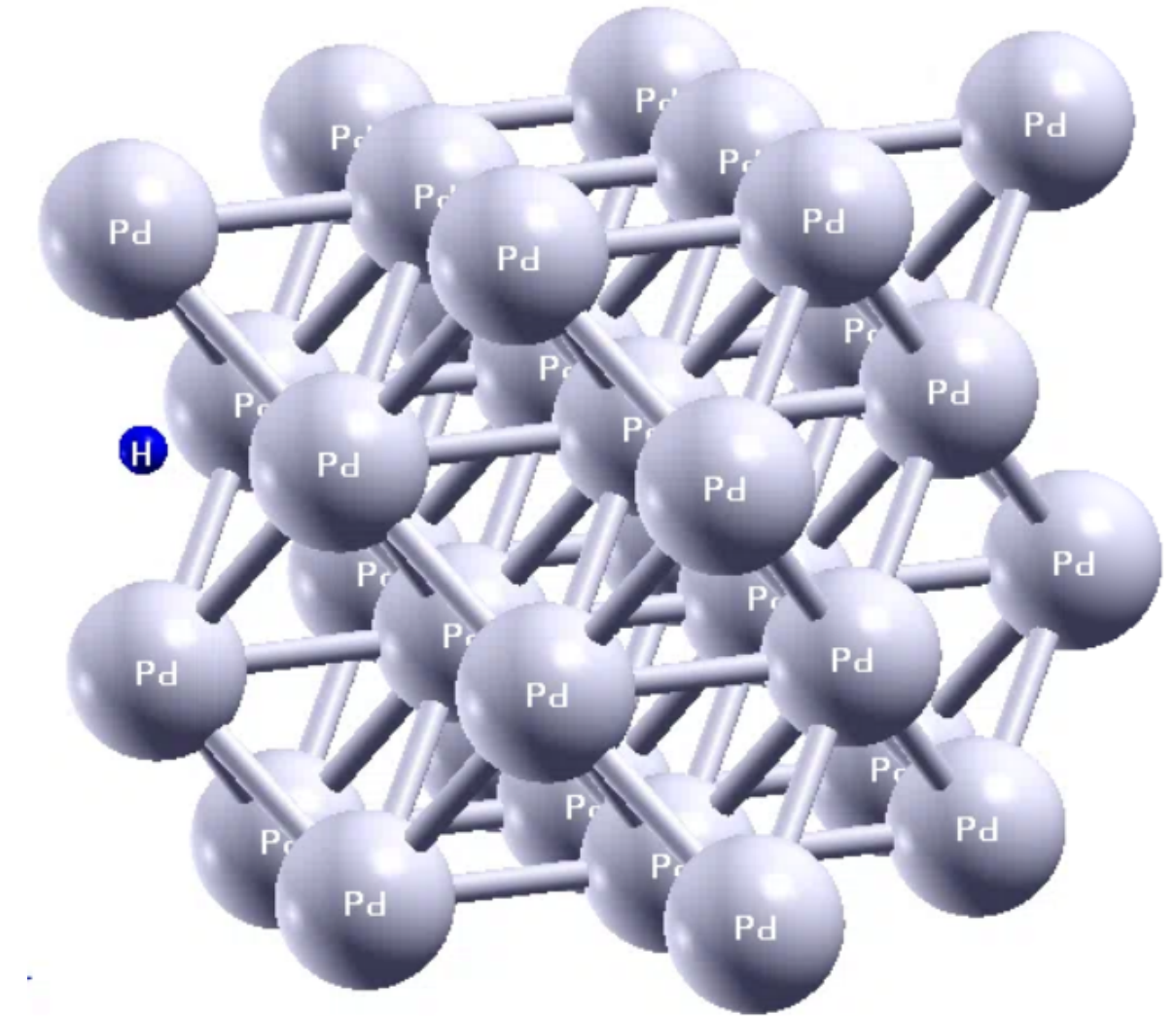
Stable Configurations of Hydrogen in Palladium

- Insert one hydrogen atom into supercell
- Symmetry: choice of voids is arbitrary
- Interstitial Voids: Hydrogen preferentially occupies octahedral and tetrahedral voids in the fcc structure

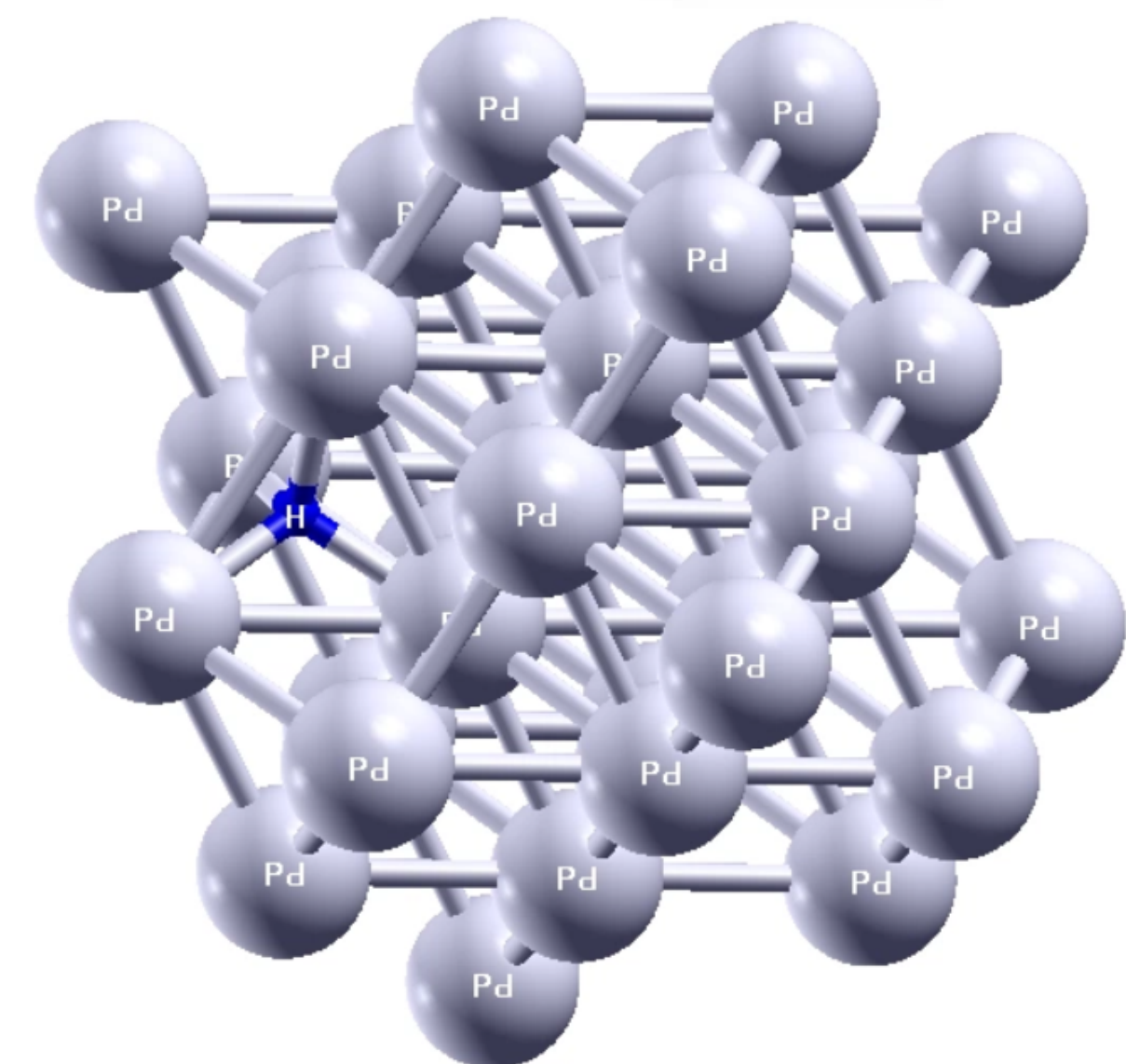
⇒ two stable configurations of hydrogen in palladium

- Energy Minima: Relaxation calculations confirm the octahedral void is the global minimum ($E_{\text{oct}} \approx -960.8412762 \text{ H}$), while the tetrahedral void is a local minimum ($E_{\text{tet}} \approx -960.8381737 \text{ H}$)

$$\vec{O} = \left(0, 0, \frac{1}{2}\right) a_{\text{lat}}$$



$$\vec{T} = \left(\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right) a_{\text{lat}}$$

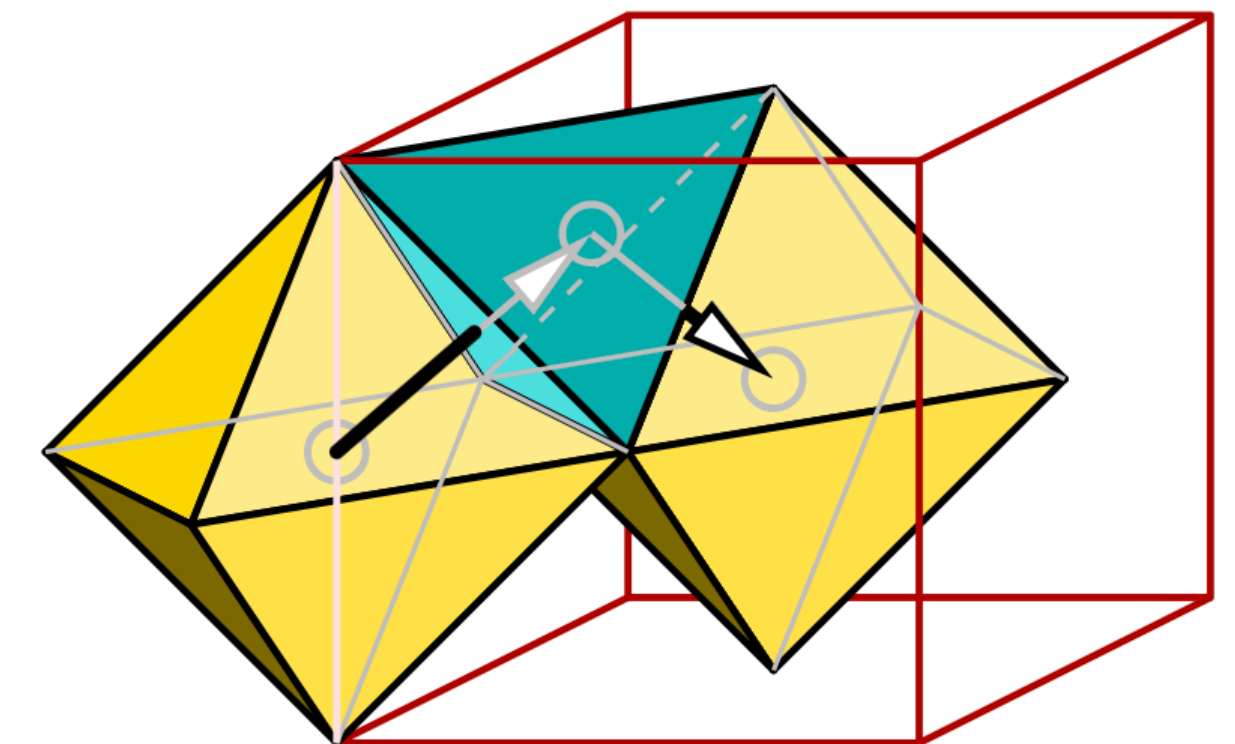
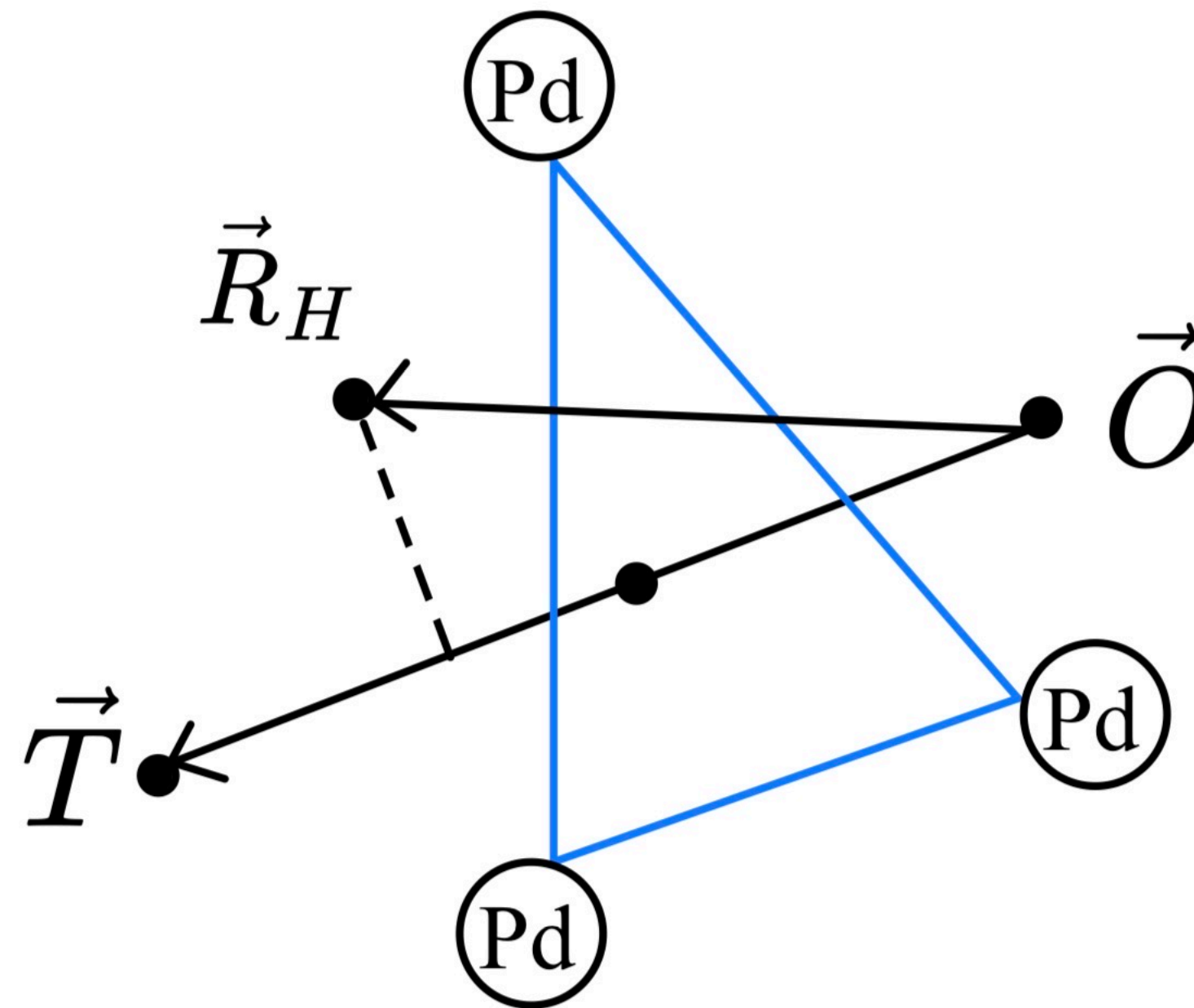


Defining the Reaction Coordinate

- Introducing dynamics: simulate transition from octahedral to tetrahedral void
- Constrain the motion of H into a plane perpendicular to the line connecting O and T
- Translation Problem: A simple constraint moving H from O to T fails because the entire crystal can artificially translate to minimize energy without actually moving the defect
- Solution: The reaction coordinate (g) must be expressed relative to the host lattice
- Lock the hydrogen atom's position relative to the center of the three specific Pd atoms forming the triangular dividing plane

$$g(R) = \frac{(\vec{T} - \vec{O}) \cdot (\vec{R}_H - \vec{O}_{\text{replaced}})}{(\vec{T} - \vec{O})^2}$$

$$\vec{R}_H(x) = \left[\vec{O} + x(\vec{T} - \vec{O}) \right] a_{lat}$$

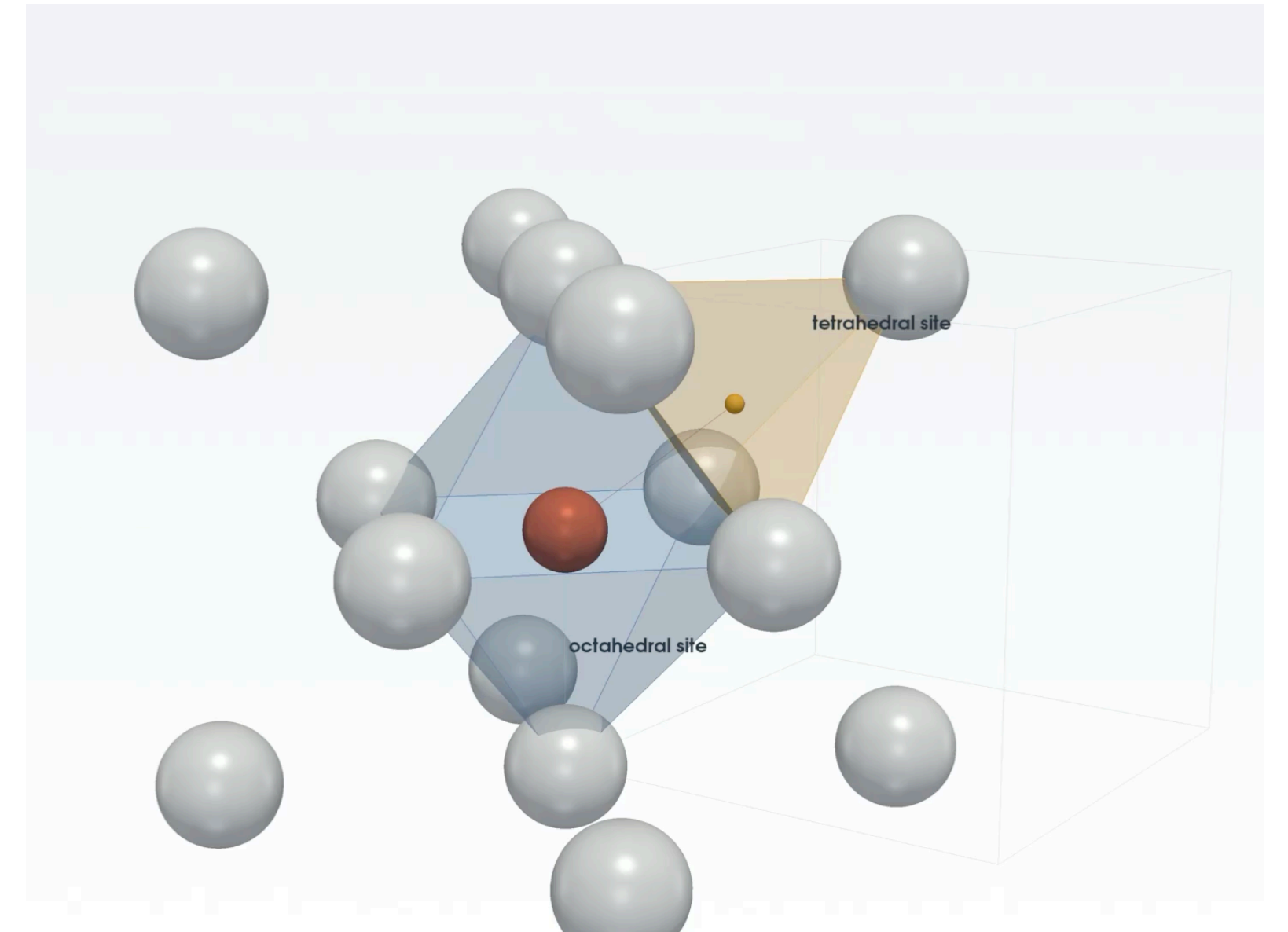


Computational Realization of the Constraint

- Expanding the projection yields a linear equation coupling the H atom to the three Pd atoms
- Shift position of H atom according to $\vec{R}_{H(x)} = \left[\vec{O} + x(\vec{T} - \vec{O}) \right] a_{lat}$, let atomic structure optimize
- $x \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$
- When the H atom is at the octahedral site, $g=0$; at the tetrahedral site, $g=1$
- CP-PAW Implementation: The !CONSTRAINT !LINEAR block applies these exact vector coefficients
- Because the sum of the atomic coefficients ($4 \times 0.444444 \approx 1.33332$ vs 3×-0.444444) equals zero, the constraint is strictly independent of global crystal translation.

$$g(\vec{R}) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \left(\frac{4}{3} \vec{R}_H - \frac{4}{9} \vec{R}_{Pd1} - \frac{4}{9} \vec{R}_{Pd2} - \frac{4}{9} \vec{R}_{Pd3} \right) + \frac{2}{3}$$

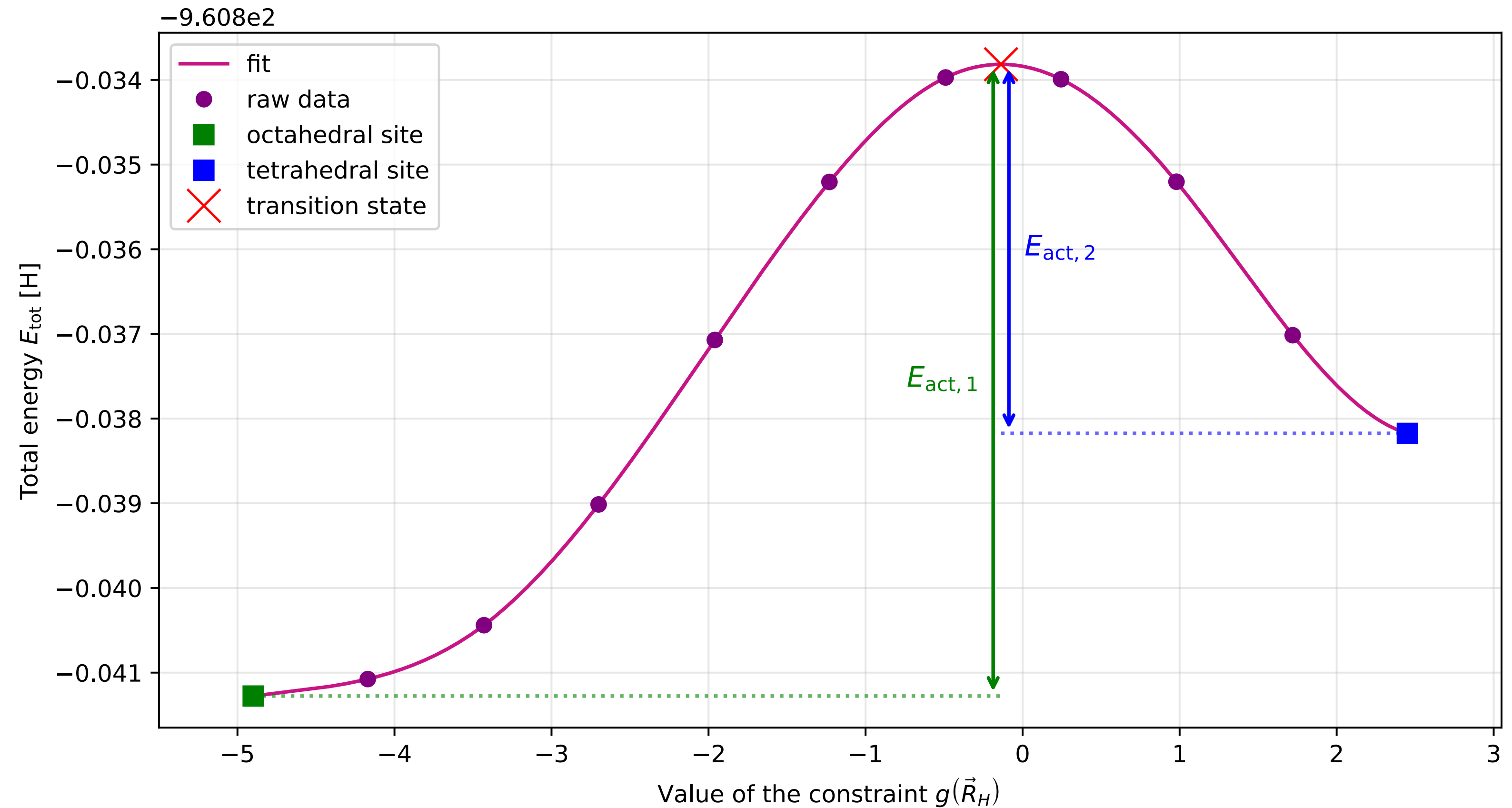
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!CONSTRAINTS
>> !LINEAR SHOW=T MOVE=F
>> !ATOM NAME='H_33' R= 1.33332 1.33332 1.33332 !END
>> !ATOM NAME='PD17' R= -0.44444 -0.44444 -0.44444 !END
>> !ATOM NAME='PD03' R= -0.44444 -0.44444 -0.44444 !END
>> !ATOM NAME='PD04' R= -0.44444 -0.44444 -0.44444 !END
>> !END
!END
```



Not! the correct path but displays the shift

Total Energy Depending on the Constraint

- Maximum of the energy profile
 $E^\ddagger \Leftrightarrow$ transition state
- $E^\ddagger = -960.8338158 \text{ H}$
- $E_{\text{act},1} = 7.46 \cdot 10^{-3} \text{ H}$
- $E_{\text{act},2} = 4.36 \cdot 10^{-3} \text{ H}$



Harmonic Approximation

- Atoms trapped inside a stable void (octahedral or tetrahedral) sit at a potential energy minimum
- $\Rightarrow$ expand the energy along the reaction coordinate using Taylor series

$$E(g) \approx E_{\min} + \frac{1}{2} \left(\frac{d^2E}{dg^2} \right)_{\min} (g - g_{\min})^2$$

- Remember: harmonic oscillator $E = \frac{1}{2}m\omega_0^2$

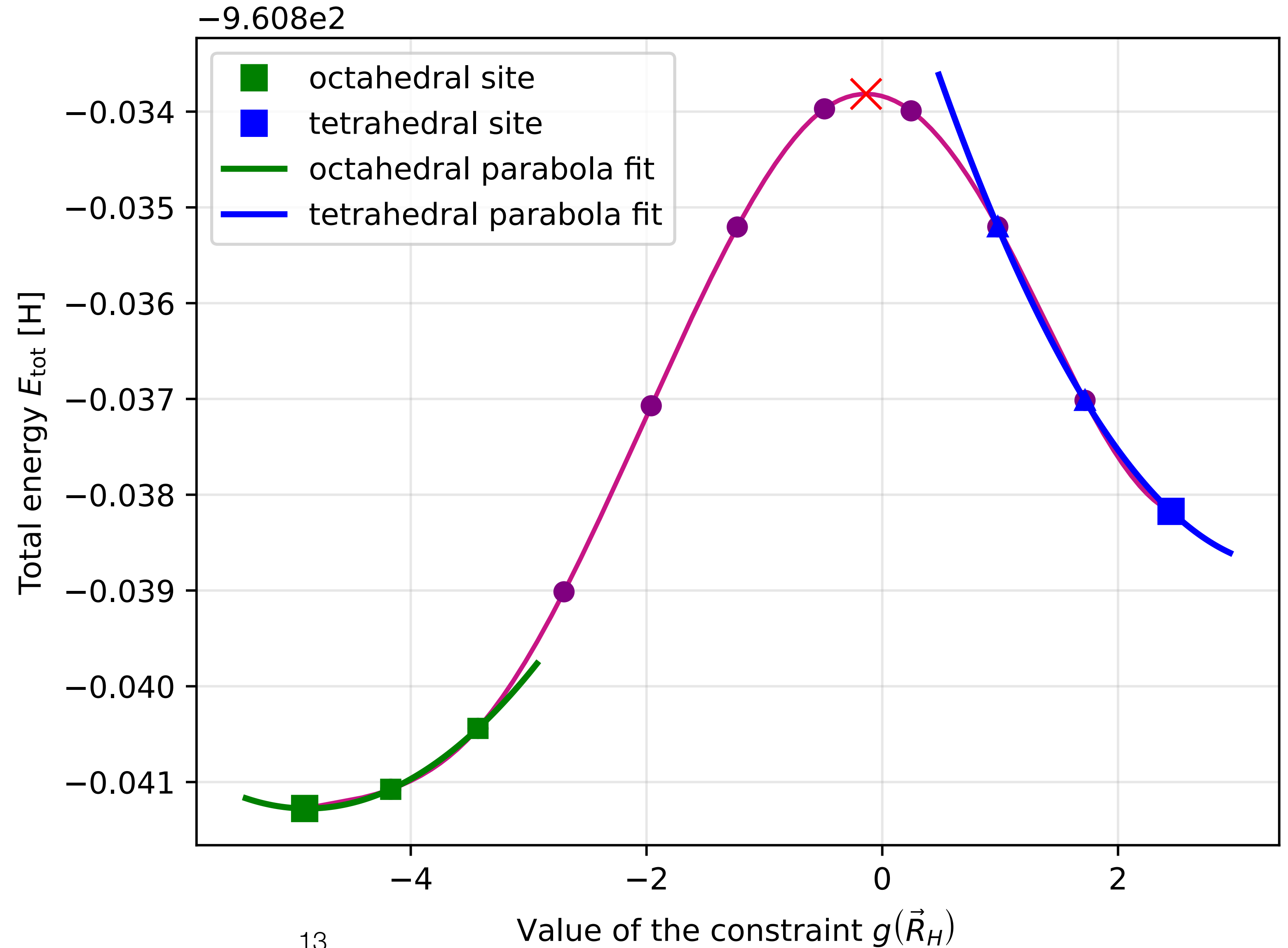
$$\left(\frac{d^2E}{dg^2} \right)_{\min} = m\omega_0^2$$

- $\Rightarrow$ Extracting the second derivative yields the attempt frequency

$$\Leftrightarrow \omega_0 = \sqrt{\frac{1}{m} \left(\frac{d^2E}{dg^2} \right)_{\min}}$$

Attempt frequencies from Curvature

- Parabolic fit of the form
 $E(g) = ag^2 + bg + c$
- Extract $a \Rightarrow m\omega_0^2 = 2a$
- Initial octahedral site:
 $\omega_{0,\text{oct}} = 2.72 \cdot 10^{12} \text{ s}^{-1}$
- Initial tetrahedral site:
 $\omega_{0,\text{tet}} = 3.3 \cdot 10^{12} \text{ s}^{-1}$



Equilibrium Probabilities

- Determine equilibrium probabilities for the hydrogen to be at the tetrahedral and octahedral site according to

$$P_i^{eq} = A \exp \left(-\frac{V_i}{k_B T} \right) \left(\frac{2\pi k_B T}{m\omega_{0,i}^2} \right)^{\frac{3}{2}}$$

- For example at room temperature $T = 298$ K: $P_{\text{oct}}^{eq} = 0.96$ and $P_{\text{tet}}^{eq} = 0.04$
- Octahedral site has lower energy and is therefore more stable than the tetrahedral site $\Rightarrow$ Hydrogen favours octahedral void

Geometric Factor

$$\sum_j \left[\left(\vec{r}_j - \vec{r}_i \right) \otimes \left(\vec{r}_j - \vec{r}_i \right) \right]$$

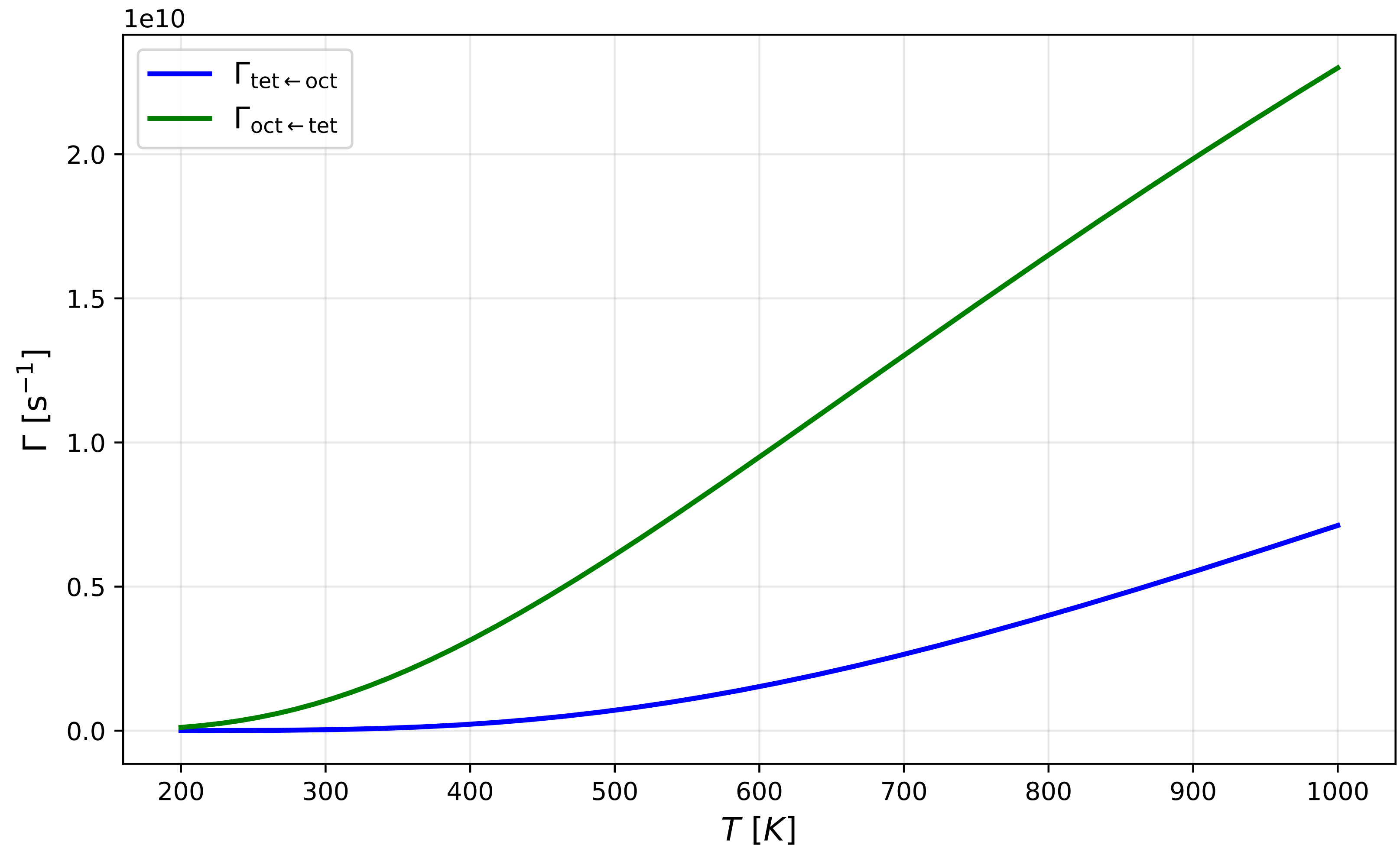
- Displacement vectors $\left(\vec{r}_j - \vec{r}_i \right)$ (shift from octahedral to tetrahedral voids and vice versa)
- 8 possibilities for jump from octahedral to tetrahedral and 4 possibilities for jump from tetrahedral to octahedral void
- Calculations $\Rightarrow$ off diagonal elements in resulting matrix vanish
- Transition Octa $\rightarrow$ Tetra $\frac{1}{2}a_{\text{lat}}^2$ and Tetra $\rightarrow$ Octa $\frac{1}{4}a_{\text{lat}}^2$

Jump Rates from Transition State Theory

- Thermally activated processes are described by TST
- Diffusion is thermally activated: hydrogen atom vibrates until energy fluctuation is sufficiently large to push it over energy barrier E_{act}
- Jump rates depend on temperature $\Gamma_{i \leftarrow j} = \frac{\omega_{0,j}}{2\pi} \exp\left(-\frac{E_{\text{act},i \leftarrow j}}{k_B T}\right)$
- Arrhenius plot: thermally activated processes form a straight line

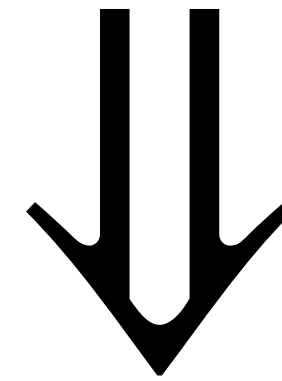
Jump Rates from Transition State Theory

- Activation energy of transition from tetrahedral to octahedral is smaller than activation energy for the inverted process
- $\Rightarrow \Gamma_{\text{tet} \leftarrow \text{oct}} < \Gamma_{\text{oct} \leftarrow \text{tet}}$
- Not shown here: Arrhenius plot is linear as expected
- Jump rates at $T = 298 \text{ K}$:
 $\Gamma_{\text{oct} \leftarrow \text{tet}} = 1.0 \cdot 10^9 \text{ s}^{-1}$
 $\Gamma_{\text{tet} \leftarrow \text{oct}} = 3.1 \cdot 10^7 \text{ s}^{-1}$



Diffusion Constant

$$D = \sum_i \frac{P_i^{\text{eq}}}{\sum_k P_k^{\text{eq}}} \sum_j \Gamma_{j \leftarrow i} \left[(\vec{r}_j - \vec{r}_i) \otimes (\vec{r}_j - \vec{r}_i) \right]$$

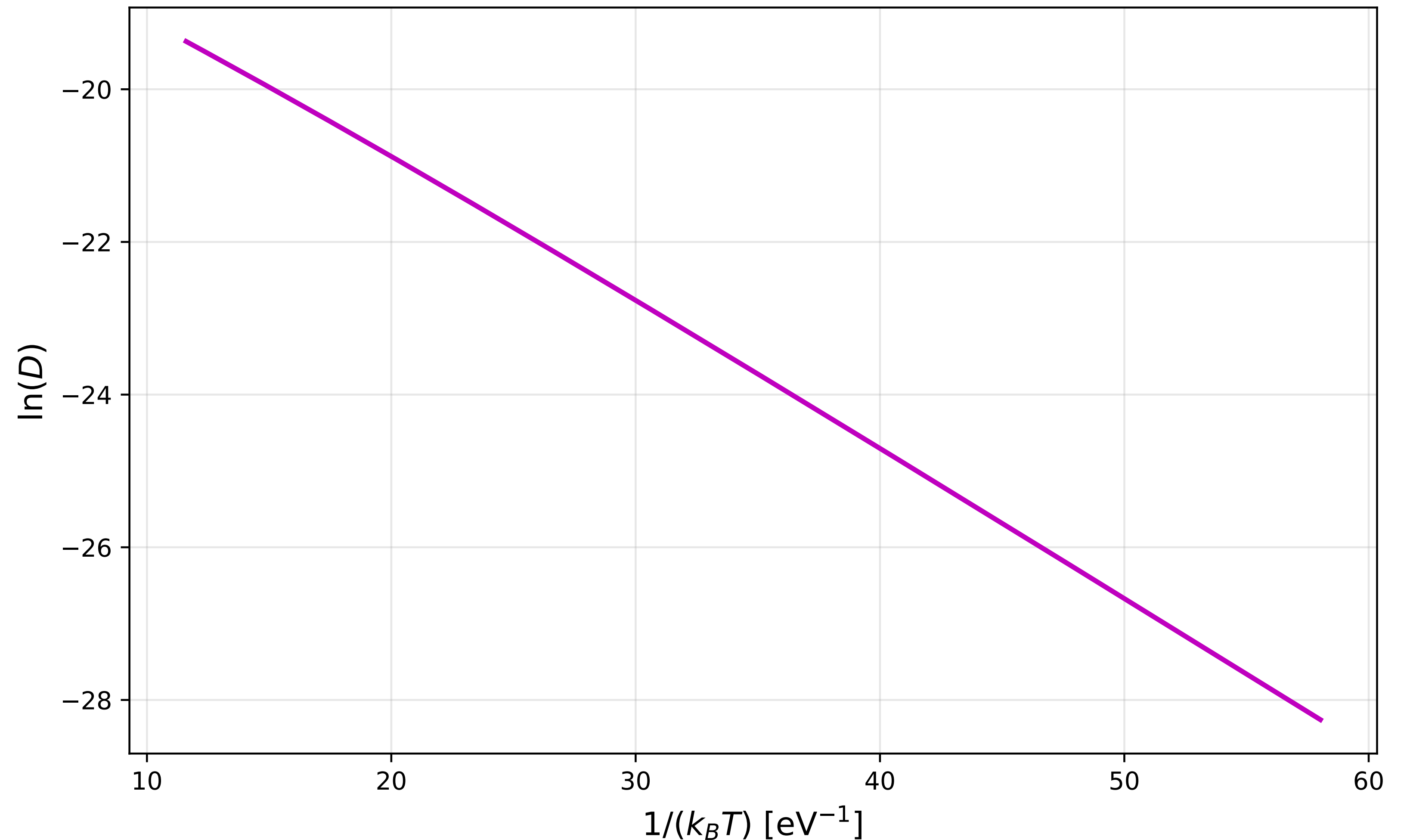


Insert everything obtained before

$$D = \frac{P_{\text{oct}}^{\text{eq}}}{\sum_k P_k^{\text{eq}}} \Gamma_{\text{tet} \leftarrow \text{oct}} \left(\frac{1}{2} a_{\text{lat}}^2 \right) + \frac{P_{\text{tet}}^{\text{eq}}}{\sum_k P_k^{\text{eq}}} \Gamma_{\text{oct} \leftarrow \text{tet}} \left(\frac{1}{4} a_{\text{lat}}^2 \right)$$

Arrhenius Plot of the Diffusion Constant

- Temperatures from 200 K to 1000 K
- Linear trend: indicates and confirms the thermally activated process which diffusion relies on
- Diffusion constant at room temperature $T=298$ K
 $D = 2.295 \cdot 10^{-11} \text{ m}^2 \text{ s}^{-1}$
- $\Rightarrow$ similar to the value provided in literature $D = 3.248 \cdot 10^{-11} \text{ m}^2 \text{ s}^{-1}$
- Deviations due to approximations, limited number of k-points etc...



Summary

- Calculated diffusion constant ($2.295 \cdot 10^{-11} \text{ m}^2 \text{ s}^{-1}$) shows sufficient agreement with the literature ($3.248 \cdot 10^{-11} \text{ m}^2 \text{ s}^{-1}$) $\Rightarrow$ The CP-PAW code qualitatively and quantitatively validates the thermally activated hopping mechanism
- Minor deviations arise from methodological constraints: Use of the harmonic approximation, a limited k-point mesh, and localized parabolic fitting
- Generally first principle calculations provide a good approximation and are sufficient to display the investigated aspects within this project

Outlook

- Increase number of k-points to obtain better results (higher computational cost!)
- Second approach for constraint: start with the actual value of the constraint and move the atomic structure such that it satisfies the constraint
- Consistency check: simulate the transition from tetrahedral to octahedral void and repeat calculations